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## ECE 4644 / 5610 Satellite Communications

### Homework Assignment #1— SOLUTIONS

**Note:** For all problems take the radius of Earth, $R_E$, to be 6378.137 km.

### Problem #1

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a. A radio transmitter is mounted on a tower 300 m tall. From geometrical considerations, determine the distance (along the curved surface of the Earth) from the base of the tower to the limit of reception. Assume straight line propagation and a surface that is featureless (except for the tower). (Hint: Sketch the situation and draw a line from the top of the tower to the point on Earth's surface where the line forms a tangent to the surface – this will define the local horizon in that direction and also a right-angled triangle with sides and angles that can be easily solved. One side of this triangle has the length $R_E$.)

b. Repeat the previous problem for a satellite at an altitude of 900 km.

c. By any means, even very rough approximation, estimate how many towers would be needed to cover the same area of Earth's surface as the satellite.

### Problem #2

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State Kepler's three laws of planetary motion. For a satellite in an elliptical orbit around the Earth, explain what is located at the two foci of the ellipse. With a sketch, explain the meaning of the terms *apogee* and *perigee* in relation to this orbit. Qualitatively, how you would expect the ratio of satellite velocity at perigee to velocity at apogee, $v_{per} / v_{apog}$, to change if an orbit is changed from circular to slightly elliptical to highly elliptical? Explain.

### Problem #3

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A satellite is in a polar circular orbit at an altitude of 600 km. Determine

a. The designation for this orbit: LEO, MEO, or GEO
b. The inclination of the orbit
c. The orbital period in minutes and seconds
d. The orbital velocity in meters per second
e. The orbital angular velocity in radians per second

**Problem #4**

The Artemis II mission will reach deep space in stages. The spacecraft will be lifted to an initial orbit with a minimum height above Earth of 184 km and a maximum height of 2,240 km. This orbit will be used to check out the spacecraft and prepare it to be launched into a higher orbit.

a. State the apogee height, $h_{apog}$ , and perigee height, $h_{per}$

b. Find the distance from the center of Earth to the point of perigee, $r_{per}$ , and the distance to the point of apogee, $r_{apog}$

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c. Determine the semi-major axis of the orbit and its eccentricity

d. Determine the period of the orbit

e. Determine the length of the semi-minor axis and the distance of the center of Earth from the center of the ellipse

f. Carefully sketch the orbit. Indicate the positions of the foci and the lengths of the semi-major and semi-minor axes. Provide an outline of Earth, centered properly and to scale. Show on the sketch the perigee height, the apogee height, the distance to perigee, and the distance to apogee

g. Comment on this initial orbit. Is there something unusual about it? What are the implications?

h. By considering Kepler's second law, determine the ratio of the satellite velocity at perigee to that at apogee, i.e., $v_{per}$ / $v_{apog}$

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PROBLEM 1

**a. A radio transmitter is mounted on a tower 300 m tall. From geometrical considerations, determine the distance (along the curved surface of the Earth) from the base of the tower to the limit of reception. Assume straight line propagation and a surface that is featureless (except for the tower). (Hint: Sketch the situation and draw a line from the top of the tower to the point on Earth's surface where the line forms a tangent to the surface – this will define the local horizon in that direction and also a right-angled triangle with sides and angles that can be easily solved. One side of this triangle has the length $R_E$.)**

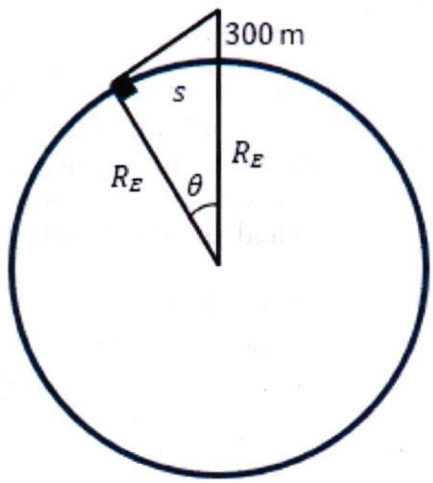


FIGURE 1. Sketch of Geometrical Considerations for Radio Transmitter Reception Limit

The triangle defined in Figure 1 can be solved for the angle $\theta$ using trigonometry:

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \longrightarrow \cos\theta = \frac{R_E}{R_E + 0.3\text{ km}}$$

$$\theta = \cos^{-1}\left(\frac{R_E}{R_E + 0.3\text{ km}}\right) = 0.0097\text{ rad}$$

The distance along the surface of the Earth $s$ is the arc length defined by the calculated angle $\theta$:

$$s_{tower} = r\theta = R_E\theta = 61.86\text{ km}$$

Therefore, the limit of reception for a radio transmitter on a 300 m tower along the surface of the Earth is $s = 61.86\text{ km}$.

**b. Repeat the previous problem for a satellite at an altitude of 900 km.**

Using 900 km instead of 0.3 km above the Earth's surface and calculating $\theta$:

$$\theta = \cos^{-1}\left(\frac{R_E}{R_E + 900\text{ km}}\right) = 0.0097\text{ rad}$$

$$s_{sat} = r\theta = R_E\theta = 3205.53\text{ km}$$

Therefore, the limit of reception for a radio transmitter on a satellite at a 900 km altitude along the surface of the Earth is $s = 3205.53\text{ km}$.

**c. By any means, even very rough approximation, estimate how many towers would be needed to cover the same area of Earth's surface as the satellite.**

Assuming that the surface of the Earth is flat rather than curved for the means of rough approximation, the area of the circle in which the tower and the satellite have coverage can be determined:

$$A_{circle} = \pi r^2$$

$$A_{tower} = \pi s_{tower}^2 = 12022.034 \text{ km}^2$$

$$A_{sat} = \pi s_{sat}^2 = 32281271.316 \text{ km}^2$$

$$\frac{A_{sat}}{A_{tower}} \approx 2685$$

The ratio of these areas reveals that approximately 2685 radio towers would be needed to cover the same area of Earth's surface.

## PROBLEM 2

**State Kepler's three laws of planetary motion. For a satellite in an elliptical orbit around the Earth, explain what is located at the two foci of the ellipse. With a sketch, explain the meaning of the terms *apogee* and *perigee* in relation to this orbit. Qualitatively, how you would expect the ratio of satellite velocity at perigee to velocity at apogee, $v_{per}$ / $v_{apog}$, to change if an orbit is changed from circular to slightly elliptical to highly elliptical? Explain.**

Kepler's three laws of planetary motion are summarized below:

(1) An orbit is defined by an ellipse, where the central body is located at one focus of the ellipse and the second focus is empty. The sketch in Figure 2 demonstrates this concept.
(2) Equal areas of the ellipse are swept out in equal durations. An example of these areas is outlined in Figure 2. If $A_1 = A_2$, then $t_{p,1} - t_{p,2} = t_{a,1} - t_{a,2}$.
(3) The orbital period is related to the semi-major axis of the orbit by the equation $T^2 = (4\pi^2 a^3)/\mu$.

For a satellite in orbit around Earth, Earth would be at one focus of the ellipse. The other focus would be empty, as defined by Kepler's first law. Perigee is the point of closest approach in an elliptical orbit of a satellite around a central body, while apogee describes the furthest point in the orbit. These points are illustrated in the sketch provided in Figure 2.

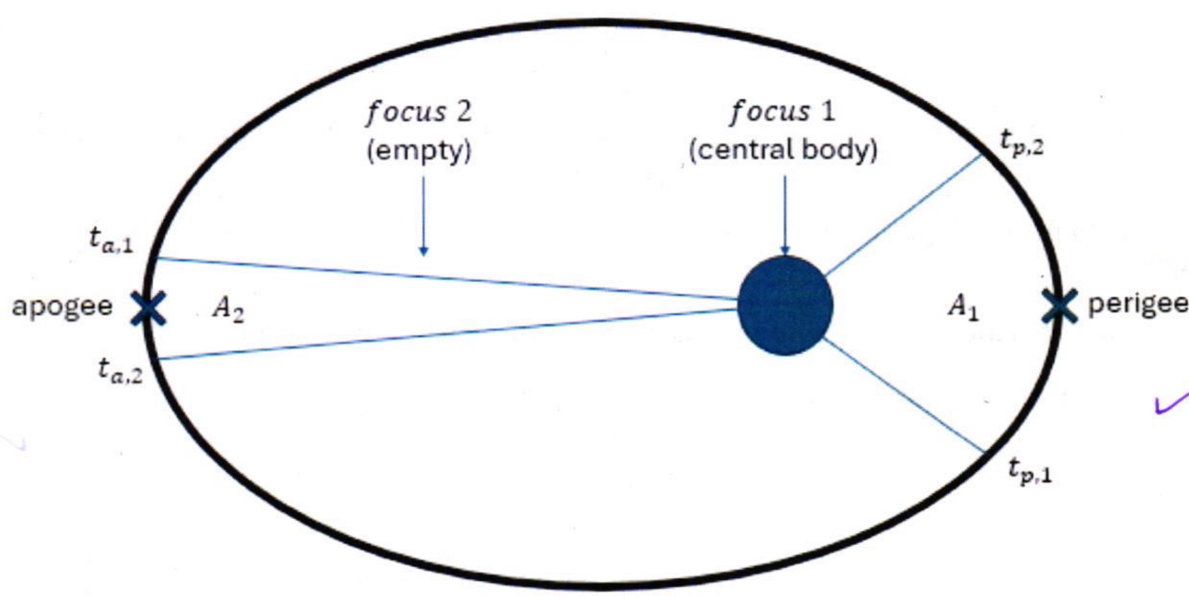


FIGURE 2. Sketch of Kepler's Laws and Essential Points

For a circular orbit, the apogee and perigee distances are not defined because the altitude in relation to the central body is constant. Therefore, the velocity is also constant, and the ratio of the instantaneous velocities $v_{per}/v_{apog}$ for any two points on the circular orbit would be equal to 1. If the orbit was changed to be slightly elliptical, the ratio $v_{per}/v_{apog}$ will be greater than 1 as the instantaneous velocity at perigee would be faster than at apogee based on Kepler's second law. This ratio will continue to get larger as the orbit in question becomes more highly elliptical. The ratio will increase with increasing eccentricity.

## Problem 3

**A satellite is in a polar circular orbit at an altitude of 600 km. Determine**

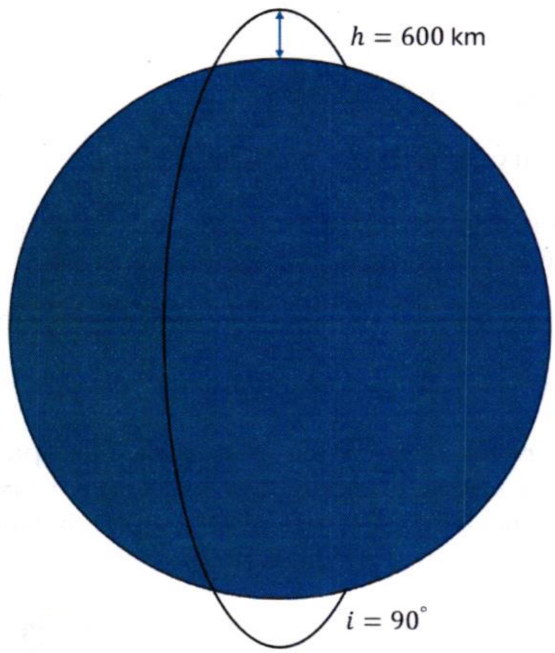


FIGURE 3. Sketch of Problem 3 Orbit

**a. The designation for this orbit: LEO, MEO, or GEO**

This orbit is $\boxed{\text{LEO}}$ because it lies within the range of approximately 250 to 1200 kilometers in altitude that defines this orbital regime.

**b. The inclination of the orbit**

A polar orbit has an inclination of $\boxed{i = 90°}$.

**c. The orbital period in minutes and seconds**

The orbital period for a circular orbit is defined by the following equation:

$$T_p = \frac{2\pi r}{v}\ ,\ \text{where } v = \sqrt{\frac{\mu}{r}}$$

Simplifying, the period of a circular orbit can be determined using given equation:

$$T_p = \frac{2\pi r^{3/2}}{\mu^{1/2}}$$

The radius of the orbit is determined by adding the height of the orbit to the radius of the Earth:

$$r = h + R_E = 600 + 6378.137 = 6978.137 \text{ km}$$

$$T_p = \frac{2\pi(6978.137)^{3/2}}{\mu^{1/2}} = 5801.23 \text{ sec}$$

The orbital period of a circular orbit at an altitude of 600 km is 5801.23 seconds or 96.6872 minutes.

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**d. The orbital velocity in meters per second**

The orbital velocity of a circular orbit is found using the equation shown:

$$v = \sqrt{\frac{\mu}{r}}$$

$$v = \sqrt{\frac{\mu}{6978.137}} = 7.558 \text{ km/s}$$

The orbital velocity is 7.5579 km/s or 7557.9 m/s.

**e. The orbital angular velocity in radians per second**

The satellite will make a complete orbit equal to $2\pi$ radians every orbital period, so the orbital angular velocity can be determined by using the previously determined orbital period.

$$\omega = \frac{2\pi}{T_P} = \frac{2\pi}{5801.23} = 0.001083 \text{ rad/s}$$

Problem 4

**The Artemis II mission will reach deep space in stages. The spacecraft will be lifted to an initial orbit with a minimum height above Earth of 184 km and a maximum height of 2,240 km. This orbit will be used to check out the spacecraft and prepare it to be launched into a higher orbit. a. State the apogee height, $h_{apog}$ , and perigee height, $h_{per}$**

The apogee height is the maximum height, therefore $h_{apog} = 2240$ km. Similarly, the perigee height is the minimum height, therefore $h_{per} = 184$ km.

**b. Find the distance from the center of Earth to the point of perigee, $r_{per}$ , and the distance to the point of apogee, $r_{apog}$**

The distance to apogee or perigee is the sum of the radius of the Earth and the altitude for the furthest and closest points, respectively:

$$r_{per} = h_{per} + R_E = 184 + 6378.137 = 6562.137 \text{ km}$$

$$r_{apog} = h_{apog} + R_E = 2240 + 6378.137 = 8618.137 \text{ km}$$

**c. Determine the semi-major axis of the orbit and its eccentricity**

The semimajor axis is calculated using the apogee and perigee distances:

$$2a = r_{per} + r_{apog} \longrightarrow a = \frac{r_{per} + r_{apog}}{2} = 7590.137 \text{ km}$$

Using the relationship between the instantaneous position of the satellite and the eccentric anomaly at the point of perigee, an expression for the eccentricity can be determined:

$$r_0 = a(1 - e\cos E) \text{ and } \cos E = 1 \longrightarrow R_E + h_{per} = a(1-e)$$

Rearranging this expression and solving for the eccentricity yields the answer to this portion of the problem:

$$e = 1 - \frac{R_E + h_{per}}{a} = 0.13544$$

**d. Determine the period of the orbit**

The equation for the orbital period from Kepler's third law is defined as shown:

$$T^2 = \frac{4\pi^2 a^3}{\mu} \longrightarrow T = \sqrt{\frac{4\pi^2 a^3}{\mu}}$$

$$T = \sqrt{\frac{4\pi^2 (7590.137)^3}{\mu}} = 6580.90 \text{ sec} = 109.682 \text{ min}$$

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**e. Determine the length of the semi-minor axis and the distance of the center of Earth from the center of the ellipse**

The semi-minor axis length can be calculated using the equation shown based on the geometry of the ellipse:

$$b = a(1 - e^2)^{1/2} = 7590.137(1 - 0.13544^2)^{1/2} = 7520.199 \text{ km}$$

The distance between the center of the Earth and the center of the ellipse is the product of the semi-major axis and the eccentricity:

$$ae = 7590.137 \times 0.13544 = 1028 \text{ km}$$

**f. Carefully sketch the orbit. Indicate the positions of the foci and the lengths of the semi-major and semi-minor axes. Provide an outline of Earth, centered properly and to scale. Show on the sketch the perigee height, the apogee height, the distance to perigee, and the distance to apogee**

A sketch of the orbit is shown in Figure 4 (see next page).

**g. Comment on this initial orbit. Is there something unusual about it? What are the implications?**

The orbit is elliptical. It is unusual in that is has a very low perigee altitude relative to what we have learned about typical values for altitude in the LEO regime, as well as an apogee value similar above traditional LEO. Since this is the first crewed mission in a significant period of time, this initial orbit passing very near to Earth at perigee might provide an opportunity to abort for the safety of the astronauts onboard if there is an issue identified early in the mission.

**h. By considering Kepler's second law, determine the ratio of the satellite velocity at perigee to that at apogee, i.e.,** $v_{per}$ / $v_{apog}$

As shown in Figure 5, the relationship between equal areas swept in equal time as defined in Kepler's second law can be approximated using triangles, where the base of the triangle is the distance at perigee and apogee respectively. The area of that triangle is calculated as shown below:

$$dA = \frac{1}{2} v \text{ dt } r$$

This expression can be developed for both the perigee and apogee areas:

$$dA_{per} = \frac{1}{2} v_{per} \text{ dt } r_{per}$$

$$dA_{apog} = \frac{1}{2} v_{apog} \text{ dt } r_{apog}$$

As Kepler's second law states that these areas must be equal over equal time, the expressions can be set equal to each other:

$$\frac{dA}{dt} = \frac{dA_{per}}{dt} = \frac{dA_{apog}}{dt}$$

Substituting in the expressions for the area of the triangles and simplifying yields an inverse relationship between the velocity and position:

$$\frac{1}{2}v_{per}r_{per} = \frac{1}{2}v_{apog}r_{apog}$$
$$v_{per}r_{per} = v_{apog}r_{apog}$$
$$\frac{v_{per}}{v_{apog}} = \frac{r_{apog}}{r_{per}}$$

Therefore, the ratio of the perigee velocity to apogee velocity can be calculated using the known values for perigee and apogee distances:

$$\frac{v_{per}}{v_{apog}} = \frac{8618.137}{6562.137} = 1.3133$$

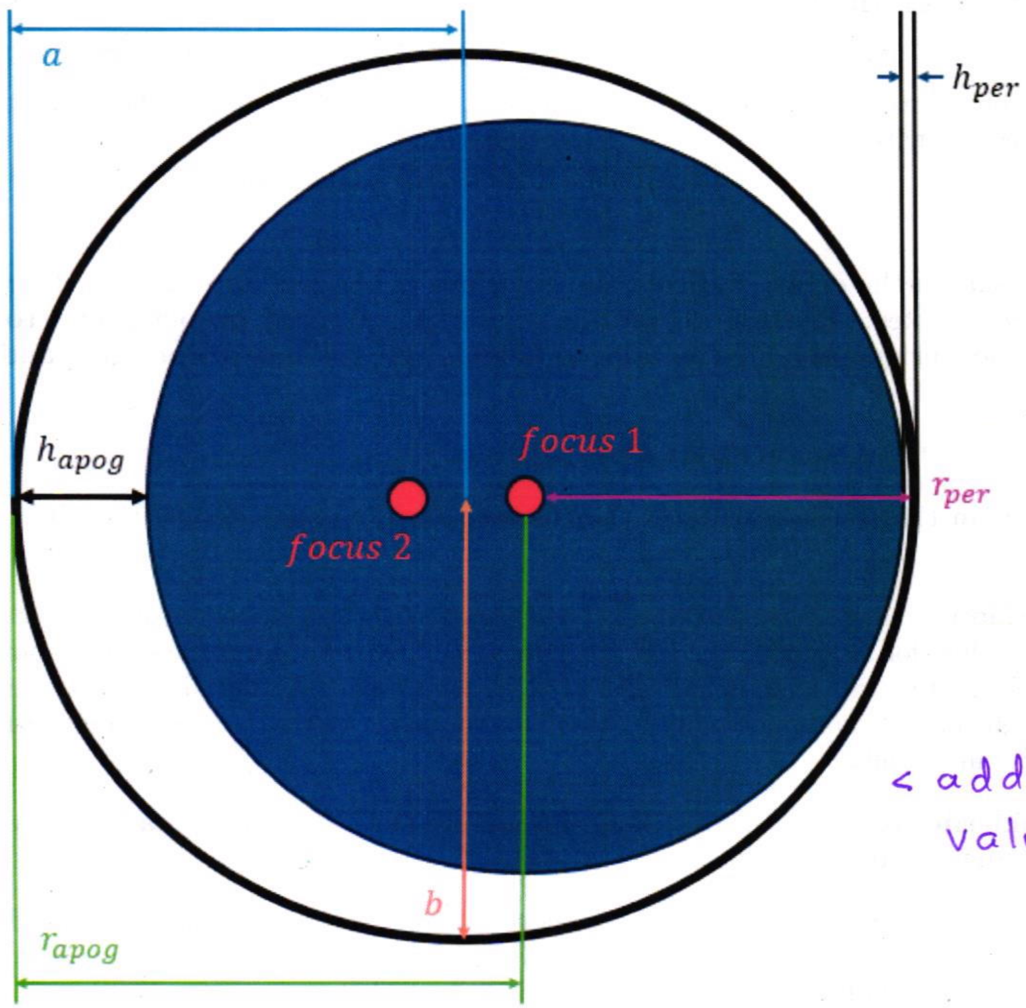


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FIGURE 4. Sketch of Initial Artemis II Orbit

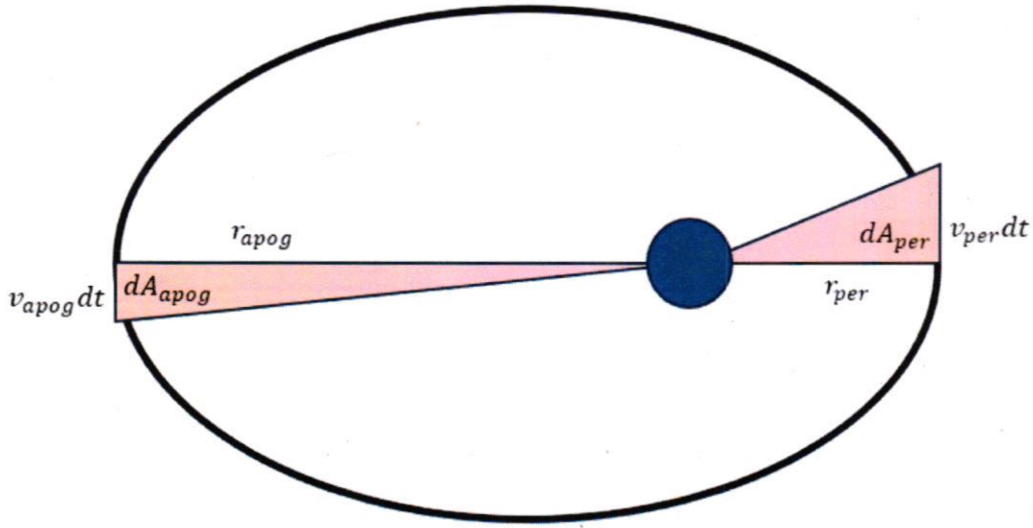


FIGURE 5. Kepler's Second Law: Equal Areas in Equal Time